Steering with Graded SAE Latent Activations

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Abstract

TODO

I'm editing in VS code but will update the overleaf periodically, then write up properly on here closer to the time!

Anonymised code is available here: https://anonymous.4open.science/r/graded-latents-70FA/

1. Introduction

Sparse Autoencoders (SAEs) (Bricken et al., 2023; Huben et al., 2024) and their variants (Gao et al., 2025; Bussmann et al., 2024; Rajamanoharan et al., 2024b; Leask et al., 2025b; Costa et al., 2025; Bussmann et al., 2025) are popular tools for recovering features from dense internal activations in language models. Ideally, SAEs faithfully approximate an activation using sparse, linear combinations of independent feature vectors from an overcomplete dictionary (Bricken et al., 2023; Karvonen et al., 2025). However, since SAEs are an unsupervised dictionary learning algorithm, they are trained using proxy metrics such as sparsity and reconstruction fidelity (Huben et al., 2024).

Substantial recent work has been focused on validating whether this training produces robust dictionaries. For example, Chanin et al. (2026) find that using sparsity as a proxy can cause feature absorption and Leask et al. (2025a) find that SAEs do not find a unique and complete dictionary of atomic features.

However, Farrell et al. (2025) present a study of an attention-only transformer model trained on an ordered bigram copying task that presents new problems for SAEs even if they recover a correct dictionary, which potentially cuts across the previous debate. They identify a novel 'relative magnitude-based' feature composition mechanism, in which both bigram orders, (x, y) and (y, x), are represented in a single fixed-length activation vector using the same basis vectors, and they are differentiated by the relative magnitude of the basis vectors (defined by scalar attention weights). They predict that an SAE trained on this activation will learn graded latent activations: a latent for each basis vector with magnitudes equal to the attention weights. In this way, an SAE could learn the correct dictionary and still provide an incomplete representation of the activation. That is, we cannot achieve a complete understanding of the activation if we assume SAE latents are independently interpretable. If this generalises to real-world language models, this could be problematic for interpretability using SAEs which commonly assumes latents are independent (McDougall et al., 2025; Karvonen et al., 2025).

However, Farrell et al. (2025) do not verify their SAE prediction. In this paper, we train a suite of BatchTopK (Bussmann et al., 2024), JumpReLU (Rajamanoharan et al., 2024b) and Matryoshka (Bussmann et al., 2025) SAEs on the toy model in Farrell et al. (2025) and provide in-depth analysis of the learned SAE behaviours. Through this, we aim to set the stage for further analysis on real-world pretrained SAEs such as Gemma Scope 1 (Lieberum et al., 2024) and 2 (McDougall et al., 2025) which use JumpReLU and Matryoshka SAEs respectively.

Our paper is structured as follows. Section 3 outlines our SAE training and evaluation methodology. In Section 4, we analyse a specific BatchTopK SAE in-depth and find that it learned two latents with graded activations that can be steered and patched into the model downstream to swap the model output from (x, y) to (y, x). This provides causal evidence that the relative magnitude-based composition mechanism ... TODO .... Moreover, these graded latents can be reliably identified by checking for anomalous positive or negative correlation with the difference between two first-layer attention weights. We check for generalisation to other SAE configurations, including JumpReLU and Matryoshka SAEs, using an automated pipeline that, given an SAE, identifies latents of interest using correlations as described, and steers them to swap outputs. Across our sweep of 1600 SAEs we find we can reliably swap between 25-70% of the transformer's original outputs, which is significant because ... TODO .... We provide our pipeline to make it easy for other researchers to train SAEs and test them similarly.

Our paper is intended to provide a foundation for future

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work searching for graded latent activations in pretrained SAEs for LLMs. Searching for these systematically in large SAEs is nontrivial. By providing avenues for automatic detection, we aim to foster this work.

2. Related Work

Sparse Autoencoders. Models can store more sparse features than the dimension of their activations naively allows through superposition, creating challenges for interpreting these models (Elhage et al., 2022). Sparse Autoencoders (SAEs) (Bricken et al., 2023; Huben et al., 2024) and their variants (Gao et al., 2025; Bussmann et al., 2024; Rajamanoharan et al., 2024b; Leask et al., 2025b; Costa et al., 2025; Bussmann et al., 2025) have been proposed as a tool for recovering these sparse features from dense activations. An SAE is typically implemented as an autoencoder with a single hidden layer, trained to reconstruct model activations while keeping the hidden representation sparse using a penalty (such as an L_1 regularization term) (Huben et al., 2024; Bricken et al., 2023). The encoder maps dense activation to a high-dimensional sparse latent space, and the decoder reconstructs the original activations using a learned dictionary of monosemantic latents, which ideally corresponds to interpretable features.

Several architectural variants address limitations in the standard ReLU-based SAE For example, standard L_1 penalties often cause "shrinkage," where the reconstructed feature magnitudes are systematically underestimated (Ferrando et al., 2024). Gated SAEs (Rajamanoharan et al., 2024a) address this by decoupling feature detection from magnitude estimation using a gated mechanism. JumpReLU SAEs (Rajamanoharan et al., 2024b) replace the standard ReLU with a discontinuous step function at a learned threshold, optimizing an L_0 penalty to achieve the same decoupling. TopK SAEs (Gao et al., 2025) replace the sparsity penalty by explicitly retaining only the k highest pre-activations per token. BatchTopK SAEs (Bussmann et al., 2024) extend this by enforcing the TopK constraint across an entire batch, allowing natural variation in the number of active latents per token while maintaining the target sparsity in expectation. Matryoshka SAEs (Bussmann et al., 2025) train nested sub-dictionaries of increasing size under a shared encoder and decoder, producing representations at multiple levels of granularity from a single training run. We use Matryoshka, BatchTopK and JumpReLU variants in this paper.

SAE performance is typically evaluated as the pareto frontier of two metrics: L_0 norm of the SAE latent activations vector, which measures sparsity, and reconstruction quality (explained variance, cross-entropy increase when patching reconstructions into the model) (Karvonen et al., 2025; Ferrando et al., 2024; Bricken et al., 2023).

SAEs have been successfully used on exploratory interpretability tasks like model auditing (Marks et al., 2025) and hypothesis generation (Movva et al., 2025), but it is unclear whether they can be used to recover the true features of models (Leask et al., 2025a; Chanin et al., 2026). For example, Leask et al. (2025a) show that SAEs can achieve high reconstruction quality while learning latents that do not correspond to any interpretable unit of computation. Therefore, reconstruction metrics are necessary but not sufficient evidence that an SAE has recovered the model's true representational structure.

SAE latents are generally treated as binary features in the literature (Paulo et al., 2025; Quirke et al., 2025), where they are 'on' if the feature is active and 'off' otherwise. In contrast, our results demonstrate graded latent activations in which relative activation magnitude encodes relational information, such as sequence order. If these results generalise to LLMs, then they could undermine this binary perspective. Additionally, graded latent activations may be ignored or misinterpreted by any downstream interpretability tool that assumes all features are independently interpretable and linear.

Relative Magnitude-based Composition. (Farrell et al., 2025) train a two-layer attention-only transformer to solve an ordered bigram copying task, in which the model must encode bigram (d_1, d_2) into a single separator token and then reconstruct the bigram in the same order as output. They find that the model learns to represent the ordered bigram at the separator token using the relative magnitude of scalar attention weights, rather than representing each possible ordering as a different direction in activation space. Specifically, they calculate the residual activation at the separator token $\boldsymbol{r}_s^{L_1}$ as:

$$\boldsymbol{r}_s^{L_1} = \alpha_{s\to d_1}(\boldsymbol{E}_{d_1}+\boldsymbol{P}_{d_1})+\alpha_{s\to d_2}(\boldsymbol{E}_{d_2}+\boldsymbol{P}_{d_2})+\boldsymbol{E}_s+\boldsymbol{P}_s \quad (1)$$

where $\alpha_{s\to d_i}$ is the attention weight in the first layer for query s (separator token) and key d_i (element of input bigram), and $\boldsymbol{E}_x$ and $\boldsymbol{P}_x$ are the token and positional embeddings for token x respectively. They predict that an SAE trained on $\boldsymbol{r}_s^{L_1}$ will learn latents corresponding to $(\boldsymbol{E}_a+\boldsymbol{P}_{d_1})$ and $(\boldsymbol{E}_b+\boldsymbol{P}_{d_2})$ with activations equal to the attention probabilities α, and that this results in graded latent activations.

Graded latent activations undermine the assumption that SAE latents are independent and binary () because ...

Steering. TODO

3. Methodology

Farrell et al. (2025) demonstrate the relative magnitude-based composition at the separator's residual activation after

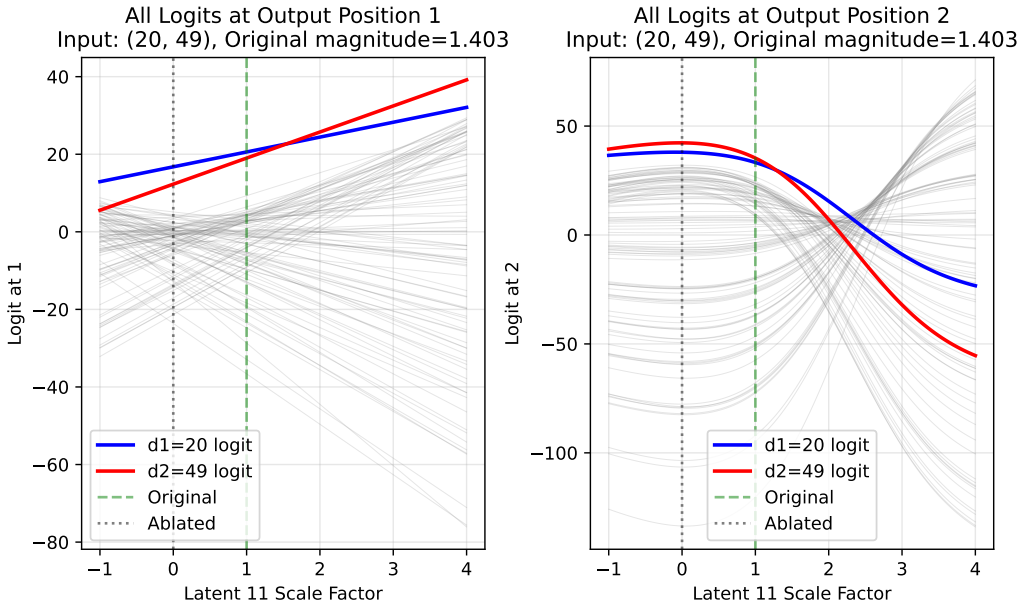

Figure 1. This graph shows for input $(20, 49)$ how the logits at the two output positions change as latent 11's activation magnitude is scaled. Each line shows the logit for some $i \in [0, 99]$ (100 lines total), such that if the top line at some latent scale represents symbol i, then i is predicted at that output position when the latent's activation magnitude is scaled by that factor. The line corresponding to the symbol in position d_1 is blue, and the symbol in position d_2 is red. For the outputs to swap from $(20, 49)$ (correct) to $(49, 20)$ by scaling the latent's activation magnitude, there must be a range of scales where the red line is on top on the left plot and the blue line is on top on the right plot. For this example, this range is $[0.16, 2.20]$.

the first layer. We refer to this vector as $\boldsymbol{r}_s^{L_1}$, where s is the separator token position, and train SAEs on that activation.

We consider 3 SAE variants: BatchTopK (Bussmann et al., 2024), JumpReLU (Rajamanoharan et al., 2024b) and Matryoshka (Bussmann et al., 2025), with implementations from Marks et al. (2024). These represent the state of the art: Matryoshka and JumpReLU SAEs are used in Gemma Scope to interpret Gemma 2 (Lieberum et al., 2024) and 3 (McDougall et al., 2025), and all three are highly cited.

We select a single BatchTopK SAE for in-depth study but validate key findings across 1,630 diverse SAE configurations and seeds (Table 1), as well as across different SAE architectures (BatchTopK, JumpReLU, Matryoshka), demonstrating that our results are not an artefact of this specific checkpoint. We select BatchTopK for its simplicity: it works by keeping the top $n \times k$ activations over a batch of n inputs and sets the rest to zero, where k is a tuned hyperparameter. This forces an average of k latent activations per input, so the actual L_0 sparsity is approximately k. In contrast, Matryoshka SAEs work similarly but force a hierarchical dictionary, and JumpReLU SAEs use the more complex JumpReLU activation function. However, by generating hypotheses based on the in-depth BatchTopK analysis and then testing them against the more complex variants, we support generalisation of our results.

To select a specific BatchTopK SAE checkpoint to analyse, we run a grid-search over $k \in [1, 5]$ and SAE expansion factors $i \in [2, 5]$, such that $d_{\text{sae}} = i \times d_{\text{model}}$. Results are shown in Table 6. We evaluate SAEs based on:

Explained variance (EV): variance of the input explained by the SAE's reconstruction is a popular and robust measure for the SAE's reconstruction quality (Bussmann et al., 2024; McDougall et al., 2025).

Patched cross-entropy loss (PCE): cross-entropy loss when patching the SAE-reconstructed activations back into the model. This is an effective measure for how the SAE impacts downstream performance, because perfect SAE reconstructions should solve the task as effectively as the model. As this reflects reconstruction quality, it is a complementary metric to EV. The baseline model cross-entropy loss was 0.1115, so PCE $= 0.1115$ is optimal.

Sparsity (L_0).

Percentage of dead (inactive) latents.

Farrell et al. (2025) train the transformer using a dataset of 10,000 bigrams: (d_1, d_2) where both elements are uniformly sampled from the range $[0, 99]$. We cache the activations at $\boldsymbol{r}_s^{L_1}$ for all bigrams.

Our grid-search shows that BatchTopK SAEs with $k = 1$ and $k = 2$ perform much worse than other SAEs for both PCE and EV. BatchTopK SAEs with $k \in [3, 5]$ are

Table 1. SAE hyperparameter sweep configurations. Braces denote swept values; fixed values are listed directly. Fixed parameters across all runs: 50,000 training steps, 1,000 warmup steps, 4,096 batch size. Total run counts are the product of all swept values and seeds. Note that $n_{\text{groups}} = 1$ in a Matryoshka SAE reduces it to a standard BatchTopK SAE, so this serves as an internal consistency check.

Parameters	**Swept Values**
d_{SAE}	$\{128, 192, 256, 320\}$
k / Target L_0 (All)	$\{3, 4, 5\}$
k / Target L_0 (BatchTopK extra)	$\{1, 2\}$
k / Target L_0 (JumpReLU extra)	$\{2\}$
JumpReLU sparsity penalty	$\{0.1, 0.5, 1.0, 5.0\}$
JumpReLU learning rate	$\{7\times10^{-5}, 3\times10^{-4}\}$
BatchTopK, Matryoshka learning rate	3×10^{-4}
Matryoshka n_{groups}	$\{1, 2, 4\}$
	Seeds / Total Runs
BatchTopK	15 seeds / 450 runs
JumpReLU	5 seeds / 640 runs
Matryoshka	15 seeds / 540 runs

joint-best for EV up to 3 s.f. and PCE up to 2 s.f. We take $k = 3$ because performance difference is negligible and lower sparsity typically improves interpretability and analysis: with at most 3 latents active per token, we can more easily exhaustively analyse each latent's role in any given computation. SAEs with $d_{\text{sae}} \in [128, 320]$ perform best for EV and PCE.

Table 2. Selected SAE configuration. The optimiser, betas, schedule and auxiliary loss are fixed in the implementation (Marks et al., 2024), and so are excluded here for brevity.

Parameter	**Value**
SAE type	BatchTopK
Input dimension (d_{model})	64
Dictionary size (d_{SAE})	128
k	3
Actual sparsity (L_0)	3.00
Learning rate	3×10^{-4}
Batch size	4096
Training steps	50,000
Seed	1
Hardware	NVIDIA GeForce RTX 2080 Ti

We therefore study a BatchTopK SAE with $k = 3$ and $d_{\text{SAE}} = 128$, as shown in Table 2. Among the trained checkpoints with this configuration, we select the one with the fewest dead latents (3.1%), which is far below the $k = 3, d_{\text{SAE}} = 128$ group average of $10.9 \pm 5.3\%$, and achieves 0.9919 EV and 0.1796 PCE. This checkpoint selection introduces selection bias toward high-performing models. We address this by confirming across 1,630 diverse SAEs that special latents (identifiable by $\Delta\alpha$ correlation) emerge reliably regardless of configuration, establishing that our findings are not specific to this checkpoint.

For analysis, we begin by inspecting max activating examples for each latent. To test if a latent has graded activations, we use steer by linearly scaling the SAE latent's activation for a given input, patch the SAE reconstruction back into the model downstream and compare the new output to the original output.

To test whether our results on the selected SAE checkpoint generalise to others, we check for similar results on 1630 different SAE configurations and seeds (Table 1) where feasible. When this is computationally infeasible, we use a targeted subset of SAE checkpoints as shown in Table 7, which were selected for strong performance and varying d_{SAE} and L_0 to provide a diverse test sample, although we note that this small set is not sufficient for statistical significance.

4. Results

4.1. Analysing Features

For the selected SAE checkpoint (Table 2) we inspect the learned latents and their activations for different inputs by plotting activation heatmaps (Figure 2). We identify two distinct types of latents that this SAE has learned (we exclude 8% of latents, which activate on fewer than 0.1% of inputs): k-symbol detectors and special latents.

k-symbol detectors (91% of latents): 88% of latents are 1-symbol detectors: they activate only if one specific symbol is present in the input at either d_1 or d_2, shown by the 'plus-sign' pattern in Figure 2. They activate most strongly where $d_1 = d_2$ (the intersection), and otherwise their activation magnitude varies only with symbol presence, not position. 3% of latents detect multiple symbols ($k > 1$) and sometimes activate with different magnitudes for different symbols (e.g., latent 5 activates stronger for symbol 7 than symbol 58), suggesting magnitude carries information.

Special latents (2 latents, 9% of activations): Only two latents (11 and 56) do not follow the above pattern. Latent 11 activates on 64% of inputs and latent 56 on 9.9%, compared to ¡4.5% for all other latents (Appendix C). Because they activate across most inputs, their activation cannot indicate whether a specific symbol is present.

Moreover, Farrell et al. (2025) find that $\boldsymbol{r}_s^{L_1}$ is a linear combination of input embeddings with attention weights as coefficients, so we compare correlations with $\Delta\alpha = \alpha_{s\to d_1} - \alpha_{s\to d_2}$, as shown in Figure 3. Latent 11's activation magnitude strongly positively correlates with $\Delta\alpha$ (Pearson correlation coefficient $r = 0.807$), and latent 56 weakly negatively correlates with $\Delta\alpha$ ($r = -0.3213$). We compute correlations over the whole dataset, and all latents have

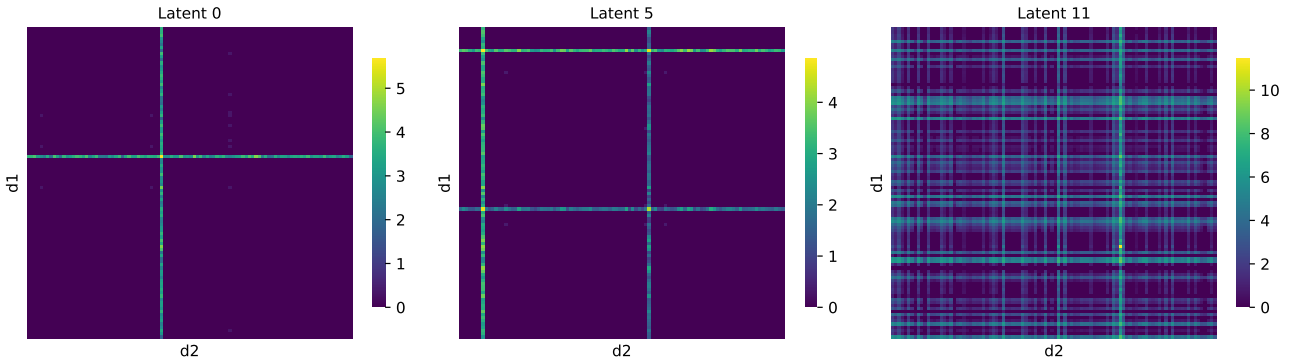

Figure 2. These 100×100 heatmaps show how specific SAE latents activate for different input bigrams (d_1, d_2), of which there are 100×100 combinations. The top-left corner cell is input $(0, 0)$ and the bottom-right corner cell is $(99, 99)$. **Left** (1-symbol detector): latent 0 activates only if d_1 or $d_2 = 41$, and activates most strongly when $d_1 = d_2 = 41$. This results in a 'plus-sign' pattern. **Center** (k-symbol detector, $k > 1$): latent 5 activates only if d_1 or d_2 is either 7 or 58, resulting in two 'plus-sign' patterns. **Right** (special latent): latent 11 activates for most inputs and activates much more strongly at peak than all other latents.

correlation $|r| < 0.004$ with $\Delta\alpha$, suggesting that these two features are indeed special relative to the others. Since latent 11 strongly positively correlates with $\Delta\alpha$ we refer to it as strongly d_1-favouring, and we similarly refer to latent 56 as weakly d_2-favouring. Latent 11 has a much stronger correlation than 56, so we refer to it as the primary special feature.

4.2. Steering Experiments

We hypothesise that special latents could have graded activations. Specifically, we hypothesise that these special latents have meaningful activation ranges corresponding to different bigram orderings. To test this hypothesis, we use steering vectors (Ferrando et al., 2024; Arditi et al., 2024) to perform a causal linear intervention: we scale special latent activations and observe whether the model's outputs change predictably.

We:

1. Input bigram (d_1, d_2) into the transformer model (Figure 5) and record the special latent's activation magnitude m for that input, using the SAE trained on $\boldsymbol{r}_s^{L_1}$.
2. Scale m by some factor p and reconstruct $\boldsymbol{r}_s^{L_1}$ as $\hat{\boldsymbol{r}}_s^{L_1}$ using the modified SAE activations.
3. Replace $\boldsymbol{r}_s^{L_1}$ with $\hat{\boldsymbol{r}}_s^{L_1}$ in the model, and compute the logits at positions (o_1, o_2) as normal.
4. Repeat for all factors $p \in [0, 10]$ with step size 0.05 across all input bigrams. Predictable and consistent shifts in the output logits under this intervention provide evidence that the special latent is graded.

Figure 1 shows a specific example input for which the predicted behaviour occurs using latent 11. When no latent scaling is applied ($p = 1$) we see that the argmax at o_1 matches the input token at d_1 as expected, and similarly o_2 matches d_2. However, for $p \in [0.16, 2.20]$ we have the opposite: o_1 has argmax d_2 and o_2 has argmax d_1. In this way, latent 11 behaves as a graded latent for this input. We define a successful output swap as follows: the symbol from d_1 is predicted at position o_2 and the symbol from d_2 is predicted at position o_1. The 'swap range' is the set of scaling factors p for which this condition holds.

We test all 10,000 input bigrams and find that steering latent 11 produces output swaps for 5,002 out of 10,000 input bigrams (50.0%). For 3,600 inputs (36%), latent 11 does not activate, so steering has no effect ($m = 0$). For the remaining 1,424 inputs (14.2%), latent 11 activates but no scaling produces a swap. This occurs for several reasons (visual examples in Appendix D):

No positive overlapping scale ranges: for 9.2% of inputs, latent 11 must be negatively scaled to swap the output at one of the output positions (Figure 7). BatchTopK SAE latent activations are strictly non-negative, where zero indicates feature absence and positive values indicate feature presence. Scaling a latent negatively does not represent absence, but rather introduces an out-of-distribution 'anti-feature' by subtracting its vector from the residual stream. We reject these cases because they force the SAE to behave in a way that is mathematically impossible during training, meaning they cannot be used as evidence for a naturally learned graded latent.

Positive but mutually exclusive scale ranges: for 2.3% of inputs, it is possible to scale the latent to swap the symbol at each output position individually but not simultaneously (Figure 8).

No swap occurs in observed ranges: for 1.6% of inputs, the logit for the symbol at d_2 is always dominant at o_2 in

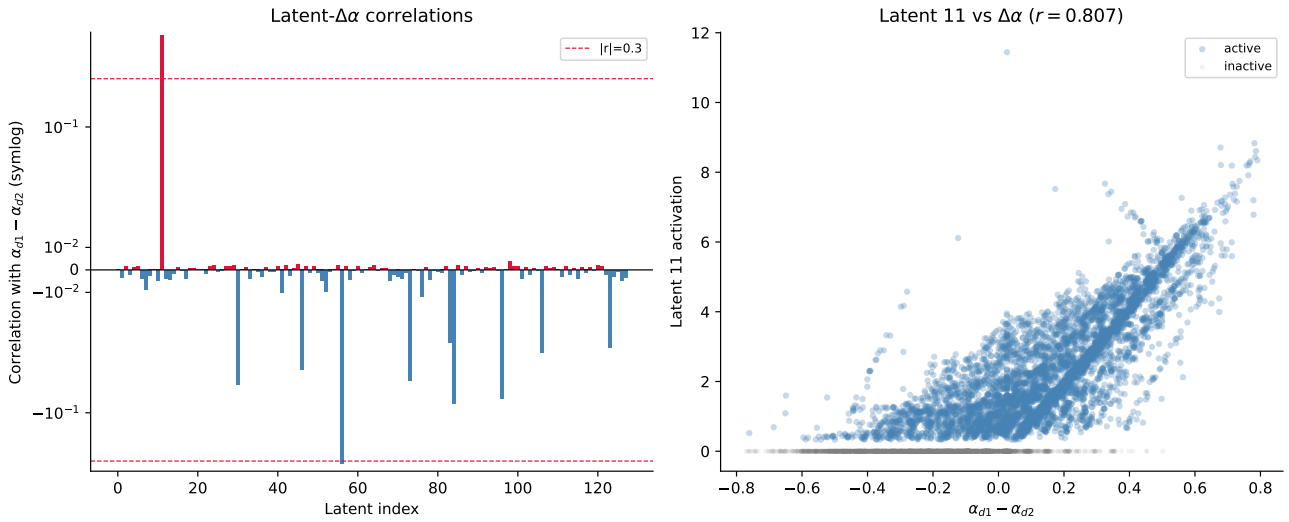

Figure 3. **Left**: This plot shows the (Pearson) correlation of each latent with $\Delta\alpha = \alpha_{s\to d_1} - \alpha_{s\to d_2}$, which is intuitively the emphasis given to d_1 over d_2 by SEP in the transformer model's first layer's attention pattern. The y-axis uses a symmetric logarithmic scale with a linear scale between $[-0.05, 0.05]$. d_1-favouring latents are highlighted red and d_2-favouring latents are highlighted blue. The top three strongest correlations are: latent 11 with $r = 0.807$, latent 56 with $r = -0.3213$, and latent 96 with $r = 0.0036$. **Right**: This plot shows the correlation of latent 11 with $\Delta\alpha$ in more detail, as this latent has the strongest correlation. All 10,000 possible input bigrams are plotted, and highlighted blue if they activate latent 11 and grey otherwise. We see that the latent activates with greater magnitude for larger $\Delta\alpha$.

the observed scale range, so there is no scale where the the symbol at d_1 is predicted instead (Figure 10). For 0.2% of inputs a similar situation occurs at o_1 (Figure 9).

Matching bigram elements: 0.8% of the inputs both activate the latent and have identical input symbols, so swapping the outputs is degenerate. The remaining 0.2% of same-symbol inputs do not activate the latent.

In limited experiments with inputs that fail for the first three causes, it is sometimes possible to scale a co-activating (non-special) latent to successfully swap outputs. For example, input $(41, 49)$ activates latents 11 (magnitude 3.1), 0 (magnitude 3.1) and 55 (magnitude 2.5). The swap fails with latent 11 because there are no positive overlapping scale ranges; however, scaling latent 55 by $p \in [1.1, 1.15]$ successfully produces a swap. This does not necessarily indicate that latent 55 is graded; rather, scaling it alters the logit distribution, which opens a valid swap zone for latent 11. Similarly, running the steering pipeline on latent 55 alone swaps only 4 inputs, all of which co-activate latent 11, suggesting a similar mechanism. We note this preliminary experiment is insufficient for statistical significance, but it suggests that scaling multiple latents together may enable more output swaps than steering special latents alone. Detailed visualisations are provided in the appendices.

We additionally steer latent 56 and successfully swap 2.0% of outputs, which is part of the 36% of inputs that do not activate latent 11 since the two special latents never coactivate.

Table 3. We perform steering experiments on several chosen and randomly sampled SAE latents to demonstrate how steering with special latents (**bold**) compares to non-special latents. We record how many outputs are successfully swapped by steering only that latent, how many do not activate the latent, and how many activate the latent but cannot be swapped.

Latent	Swapped	No Activation	Active & No Swap
0	6 (0.1%)	9786 (97.9%)	208 (2.1%)
5	23 (0.2%)	9594 (95.9%)	383 (3.8%)
11	**4976 (49.8%)**	**3600 (36.0%)**	**1424 (14.2%)**
55	4 (0.0%)	9787 (97.9%)	209 (2.1%)
56	**196 (2.0%)**	**9009 (90.1%)**	**795 (7.9%)**
63	4 (0.0%)	9801 (98.0%)	195 (2.0%)
82	4 (0.0%)	9787 (97.9%)	209 (2.1%)
117	37 (0.4%)	9782 (97.8%)	181 (1.8%)

We do not attempt steering on all latents due to computational constraints, but we attempt it on five randomly selected latents (0, 55, 63, 82, 117) and latent 96 (non-special latent with highest $\Delta\alpha$ correlation, $r = 0.0036$). Results are shown in Table 3. We find that every single successful swap using a non-special latent has a co-activating special latent, which suggests that these non-special latents are unlikely to be graded, otherwise they could singlehandedly swap more outputs, and instead alter the logit distribution such that the graded special latents can successfully swap outputs.

4.3. Generalisation to other SAEs

To confirm these findings are not a quirk of this specific SAE, we observe whether they occur across other SAEs.

To achieve this at scale, we automatically identify special latents by checking each latent's correlation with $\Delta\alpha$ is over threshold $|r| > 0.3$ (Figure 3). Figure 4 shows that SAEs with strong performance reliably learn special latents. This strongly suggests that the behaviour described so far in this section generalises to various SAE configurations. However, this generalisation is limited because we only tested three SAE types (BatchTopK, Matryoshka, JumpReLU) over 15 seeds. Future work should check for generalisation to more SAE variants such as those in Table 5, and over more training seeds for more robust statistical significance testing.

To test whether special latents can be steered to swap outputs in other SAEs, we provide the following pipeline that works given any SAE, which is available in our code repo:

1. Store the SAE activations for all input bigrams by running a single forward pass as in Figure 5 over the full dataset and recording each SAE latent's activation magnitude.
2. Identify special latents using $\Delta\alpha$ correlation $|r| > 0.3$ as a heuristic, as described previously.
3. For each identified special latent, run a batched steering grid search over all input bigrams. For each bigram and each scale factor $p \in [0, 10]$ with step size 0.05 (201 values), record the logits at output positions o_1 and o_2 under the steered reconstruction $\hat{\boldsymbol{r}}_s^{L_1}$. Bigrams for which the latent does not activate ($m = 0$) are skipped immediately since scaling has no effect.
4. Find crossover scales for each bigram:
   - For o_1: since o_1 logits are empirically linear in p, fit lines to $\ell_{d_1}(p)$ and $\ell_{d_2}(p)$ and compute the analytical intersection. Inputs for which either logit series fails an $R^2 \geq 0.999$ linearity check, or for which the intersection falls on a negative scale, are flagged and excluded.
   - For o_2: detect sign changes in $\ell_{d_1}(p) - \ell_{d_2}(p)$ on the coarse $[0, 10]$ grid, then refine each crossing to three decimal places using bisection search. All bisection tasks across the batch are executed in parallel to avoid sequential per-input forward passes.
5. Determine a valid swap zone $[p_{\text{lo}}, p_{\text{hi}}]$ per bigram by intersecting the o_1 and o_2 crossover constraints with the scale ranges over which $\text{argmax}(o_1) = d_2$ and $\text{argmax}(o_2) = d_1$ hold simultaneously. If no contiguous window satisfying both conditions exists, the bigram is recorded as a failure with a specific reason (see Section 4.2).
6. Verify each swap zone by running the model at the midpoint $p^* = (p_{\text{lo}} + p_{\text{hi}})/2$ and confirming that the model output is (d_2, d_1).
7. Report the fraction of bigrams for which a valid swap zone was found and the swap verified, along with a breakdown of failure reasons for the remainder.

Steps 3-5 of the pipeline are very computationally expensive, so we test generalisation using a varied sample of six high performing SAEs (Table 7) rather than running the pipeline on all trained SAEs (as in Figure 4). The results are shown in Table 4. We find that all six sampled SAEs learned at least one special latent ($|r| > 0.3$), and five learned a pair of opposing special latents (one d_1-favouring and one d_2-favouring). In every tested SAE, steering at least one of these special latents successfully swapped the predicted outputs for a substantial portion of the input space, ranging from 26.6% to 72.5% of all inputs. When the output swap failed, the inputs typically either did not activate the latent at all or encountered constraints similar to those detailed in Section 4.2, such as requiring negative activation scaling. This suggests that the relative magnitude-based mechanism, and the emergence of steerable, graded latent activations, generalizes consistently across varying SAE architectures and hyperparameter configurations. In future work, this study should be extended to further configurations for robust statistical significance.

5. Discussion

Our SAEs reliably learn latents with graded activations that can be steered to swap model outputs.

TODO

6. Conclusion and Future Work

In this paper, we showed that SAEs trained on an activation that uses relative magnitude-based composition reliably learn graded special latents whose activation magnitudes correlate with $\Delta\alpha$, and demonstrated that steering these latents swaps the model's predicted output for over 50% of inputs. These results indicate that the activation values of SAE latents, and not only their active/inactive status, carry information that is necessary for understanding model behaviour in this setting, which is a different perspective on transformer features than is currently prevalent in the mechanistic interpretability literature. We release our trained SAEs on Weights & Biases to support future work and reproducibility of our results.

Limitations. The base transfomer is trained on a single

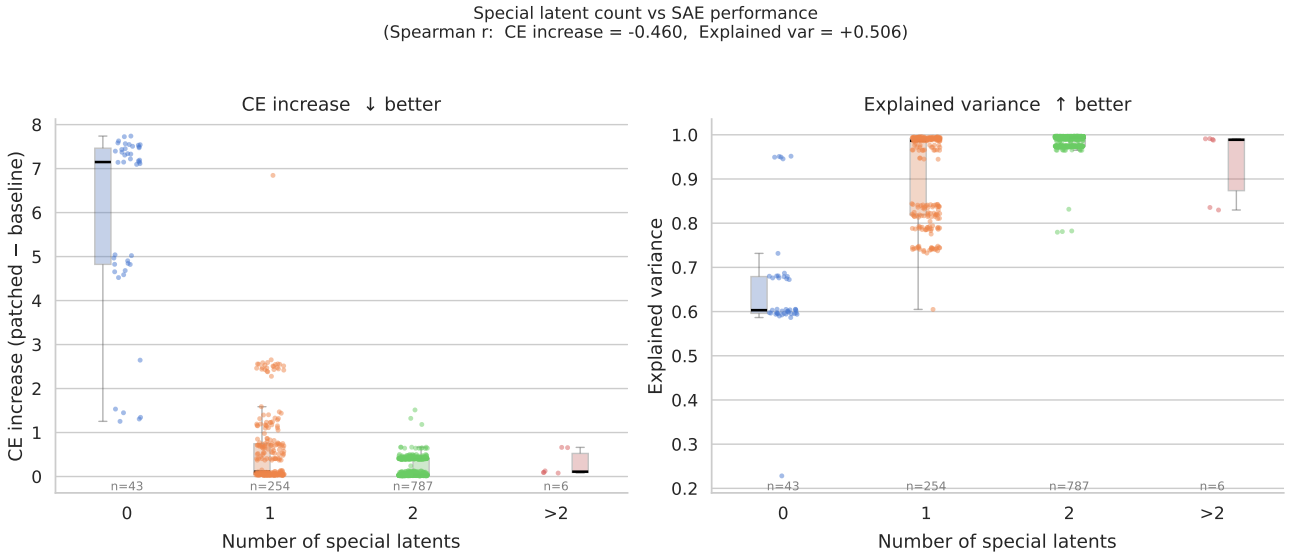

Figure 4. These strip plots show how the number of special latents (identified by the latent's activation's correlation with $\Delta\alpha$ using threshold $|r| > 0.3$, as in Section 4.1) differs across SAEs with different performance. It shows that SAEs with strong performance, measured by explained variance and patched cross-entropy loss, tend to have 2 special latents (spearman $r = -0.460$ for patched cross-entropy loss and $r = 0.506$ for explained variance).

minimal toy model with a narrow, controlled task. The architecture omits components standard in pretrained LLMs (MLPs, LayerNorm, multi-head attention, value and output projections) and uses a vocabulary of only 100 symbols. However, our SAE experiments demonstrate robust findings across diverse configurations: special latents (characterized by $\Delta\alpha$ correlation) emerge across three distinct SAE architectures (BatchTopK, JumpReLU, Matryoshka) with varying sparsity and dictionary size settings 1. This suggests the mechanism may be fundamental to how SAEs represent the feature composition in this model, though whether relative magnitude-based composition appears in less constrained models remains an open question.

Additionally, the steering pipeline currently handles one special latent at a time; simultaneous multi-latent steering could cause even more outputs to be swapped, as suggested by the co-activation experiments in Section 4.2. Finally, our SAE checkpoint was selected on the basis of having few dead latents, which introduces selection bias toward high-quality SAEs. While generalisation experiments (Section 4.3) show key findings hold broadly, a fully representative study would require analysis across random samples of all trained SAEs.

Future Work. The most important next step is determining whether pretrained LLMs implement similar mechanisms. The GemmaScope suites of SAEs (Lieberum et al., 2024; McDougall et al., 2025) provide a natural test environment, as it includes JumpReLU and Matryoshka SAEs trained on Gemma 2 and Gemma 3 at multiple widths; the same SAE variants we study here. The specific target would be identifying SAE latents in pretrained models where activation magnitude, rather than feature presence, is causally relevant for downstream behaviour.

Moreover, the steering pipeline could be extended in two directions: improving computational efficiency to allow exhaustive testing across all trained SAE configurations rather than a cherry-picked sample, and testing whether scaling multiple latents simultaneously can recover the remaining inputs for which single-latent steering fails. Preliminary evidence in Section 4.2 suggests this may be possible.

Finally, (Farrell et al., 2025) did not fully characterise the failure modes of the base transformer model (roughly 9% of inputs), and it is worth investigating whether these relate systematically to the cases where latent steering also fails.

In summary, the findings in this paper are intended as a foundation for further research on magnitude-based feature composition mechanisms in LLMs.

Impact Statement

Authors are **required** to include a statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. This statement should be in an unnumbered section at the end of the paper (co-located with Acknowledgements – the two may appear in either order, but both must be before References), and does not count toward the paper page limit. In many cases, where the ethical impacts and expected societal implications are those that are well established when advancing the field of Machine Learning, substantial discussion is not required,

Table 4. Generalisation of special latent steering to other SAE configurations from Table 7. For each identified special latent ($|r| > 0.3$), we report its bias, Pearson correlation r with $\Delta\alpha$, and the percentage of inputs where the latent was successfully steered to swap outputs (Success), did not activate (No Act.), or activated but could not be swapped (Active & No Swap).

SAE Type	d_{SAE}	L_0	Latent	Type	r	Steering Outcomes (%)		
						Success	No Act.	Active & No Swap
BatchTopK (Studied)	128	3.00	11	d_1-favouring	$+0.807$	49.8%	36.0%	14.2%
			56	d_2-favouring	-0.321	2.0%	90.1%	7.9%
BatchTopK	192	3.36	47	d_2-favouring	-0.836	52.9%	34.4%	12.7%
			116	d_1-favouring	$+0.365$	0.5%	91.7%	7.8%
	128	4.90	68	d_1-favouring	$+0.845$	32.9%	44.7%	22.4%
			64	d_2-favouring	-0.730	6.0%	62.0%	32.0%
JumpReLU	256	4.03	195	d_1-favouring	$+0.849$	38.8%	39.8%	21.4%
			179	d_2-favouring	-0.721	5.8%	60.2%	34.0%
	128	3.24	105	d_2-favouring	-0.816	28.6%	37.8%	33.6%
			4	d_1-favouring	$+0.756$	6.9%	62.3%	30.8%
Matryoshka	256	3.32	48	d_2-favouring	-0.796	26.6%	50.6%	22.8%
			62	d_1-favouring	$+0.591$	6.0%	71.2%	22.8%
	192	3.98	30	d_1-favouring	$+0.866$	72.5%	16.5%	11.0%

and a simple statement such as the following will suffice:

"This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here."

The above statement can be used verbatim in such cases, but we encourage authors to think about whether there is content which does warrant further discussion, as this statement will be apparent if the paper is later flagged for ethics review.

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A. Further comparison of SAE Types

Table 5. Comparison of some sparse autoencoder (SAE) variants.

Type	Description
Standard ReLU (Bricken et al., 2023)	Applies an L_1 penalty to hidden activations using a standard ReLU activation function to encourage sparsity.
Gated (Rajamanoharan et al., 2024a)	Decouples feature detection from magnitude estimation using a gated mechanism to prevent the shrinkage issues caused by standard L_1 penalties.
JumpReLU (Rajamanoharan et al., 2024b)	Uses a discontinuous step function at a learned per-feature threshold, approximating an L_0 penalty through straight-through estimation. Notably used in (Lieberum et al., 2024).
BatchTopK (Bussmann et al., 2024)	Retains top $n \times k$ activations over a batch of n inputs and sets the rest to zero, where k is a tuned hyperparameter. This forces an average of k latent activations per input, so the actual L_0 sparsity is approximately k. Notably used in (Brixi et al., 2026; Karvonen et al., 2025).
Matryoshka (Bussmann et al., 2025)	Trains nested sub-dictionaries of increasing size that share a single encoder/decoder, producing multiple resolutions of feature granularity in one run. Notably used in (McDougall et al., 2025).

B. Experimental Setup

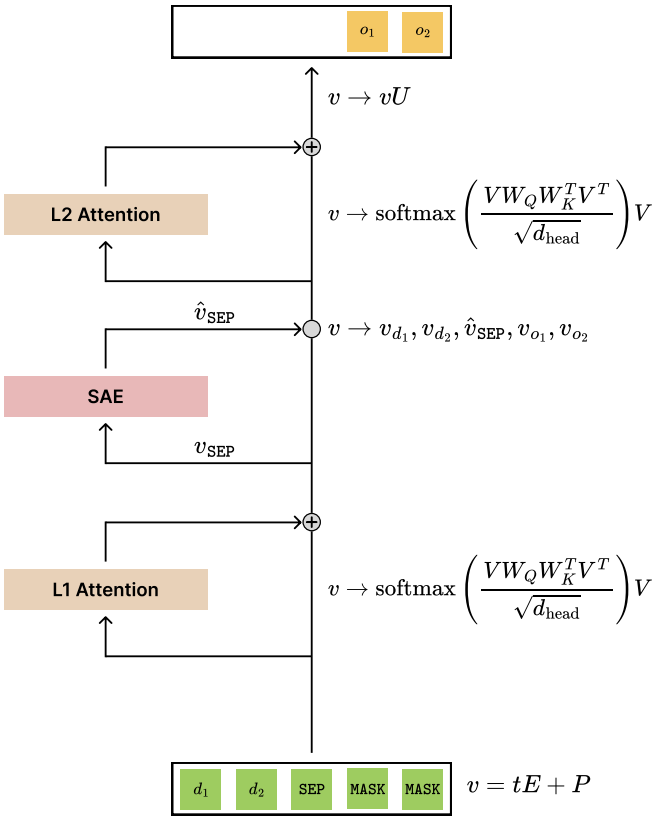

Figure 5. This diagram shows the structure of the two-layer attention-only transformer and how the SAE is used to extract and analyse the learned representations. v denotes the residual stream, which is read from and written to after each layer and has shape $(5, d_{\text{model}})$. t is the vector of input tokens $[d_1, d_2, \texttt{SEP}, \texttt{MASK}, \texttt{MASK}]$, E and P are the token and position embedding matrices respectively, U is the unembedding matrix, and each attention layer has key and query matrices, $\boldsymbol{K}^i$ and $\boldsymbol{Q}^i$ respectively, for $i \in [1, 2, 3]$. The value and output matrices for each layer are fixed to the identity, and $d_{\text{head}} = 64$. The SAE takes the residual vector at $\texttt{SEP}$, $v_{\texttt{SEP}}$, as input and patches reconstruction $\hat{v}_{\texttt{SEP}}$ back into the stream. Accuracy is calculated on the final two output logits only, represented by (o_1, o_2).

Table 6. SAE sweep results aggregated over HOW MANY [TODO] seeds and learning rates (mean ± std). L_0: mean active features per token. d_{SAE}: dictionary size. **EV** (explained variance): fraction of activation variance recovered by the SAE reconstruction (↑ better). **PCE** (patched cross-entropy): language-model CE after substituting SAE reconstructions for true activations (↓ better; baseline = 0.1115). **Dead %**: percentage of dictionary features with zero activation across the evaluation set (↓ better). **Bold**: best within each L_0 block; **underlined**: best overall.

L_0	d_{SAE}	Runs	EV (↑)	PCE (↓)	Dead % (↓)
1	128	15	0.6796 ± 0.0039	4.8800 ± 0.1734	**12.9 ± 2.9**
1	192	15	0.6755 ± 0.0036	$\mathbf{4.7661 \pm 0.1141}$	43.9 ± 1.9
1	256	15	0.6824 ± 0.0052	4.7997 ± 0.0903	53.9 ± 2.0
1	320	15	$\mathbf{0.6859 \pm 0.0039}$	4.8711 ± 0.1156	61.9 ± 1.3
2	128	14	0.9478 ± 0.0023	$\mathbf{1.5176 \pm 0.0874}$	**17.9 ± 1.3**
2	192	15	0.9461 ± 0.0034	1.6309 ± 0.2440	42.6 ± 1.4
2	256	15	0.9452 ± 0.0084	1.6028 ± 0.1765	54.4 ± 1.5
2	320	15	$\mathbf{0.9493 \pm 0.0036}$	1.6711 ± 0.1941	61.6 ± 1.8
3	128	19	0.9465 ± 0.1747	1.4374 ± 2.9323	**13.0 ± 3.9**
3	192	14	0.9940 ± 0.0005	0.1444 ± 0.0039	34.8 ± 4.8
3	256	12	0.9940 ± 0.0008	$\mathbf{0.1431 \pm 0.0040}$	48.2 ± 7.1
3	320	13	$\mathbf{0.9943 \pm 0.0008}$	0.1438 ± 0.0035	54.1 ± 6.9
4	128	15	0.9939 ± 0.0002	0.1434 ± 0.0015	**1.2 ± 0.9**
4	192	12	0.9953 ± 0.0003	0.1420 ± 0.0012	5.1 ± 2.0
4	256	15	0.9954 ± 0.0023	0.1419 ± 0.0040	17.9 ± 15.8
4	320	16	$\mathbf{0.9965 \pm 0.0018}$	$\mathbf{0.1390 \pm 0.0020}$	19.4 ± 12.6
5	128	15	0.9939 ± 0.0001	0.1417 ± 0.0013	**0.3 ± 0.4**
5	192	19	0.9946 ± 0.0021	0.1412 ± 0.0020	5.2 ± 8.3
5	256	17	0.9965 ± 0.0005	0.1399 ± 0.0016	7.2 ± 3.3
5	320	15	$\mathbf{0.9973 \pm 0.0004}$	$\mathbf{0.1382 \pm 0.0013}$	11.8 ± 3.2

C. Further SAE Latent Analysis

blabla

However, in JumpReLU test we find a latent that activates more than the specials but doesnt seem to be a special (no alpha_diff corr or scaling)

D. Further Latent Steering Examples

Table 7. Comparison of cherry-picked SAE architecture types across configuration and performance metrics. These SAEs are used to support generalisation claims when a large testing sample is not computationally feasible. Dashes indicate parameters not applicable to a given architecture. L_0 (actual) is the mean number of active latents measured at eval time. **Bold**: best value per performance metric.

SAE Type	Configuration					Performance			
	d_{SAE}	LR	k / Target L_0	Sp. Pen.	n_{groups}	CE Increase ↓	Expl. Var. ↑	Dead (%) ↓	L_0 (actual)
BatchTopK	192	3×10^{-4}	3	—	—	0.0257	0.9929	26.04	3.36
	128	3×10^{-4}	5	—	—	0.0292	0.9939	**0.78**	4.90
JumpReLU	256	7×10^{-5}	5	5	—	**0.0141**	**0.9965**	50.78	4.03
	128	7×10^{-5}	5	5	—	0.0176	0.9963	17.97	**3.24**
Matryoshka	256	3×10^{-4}	3	—	2	0.0262	0.9928	39.06	3.32
	192	3×10^{-4}	4	—	4	0.0325	0.9927	12.50	3.98

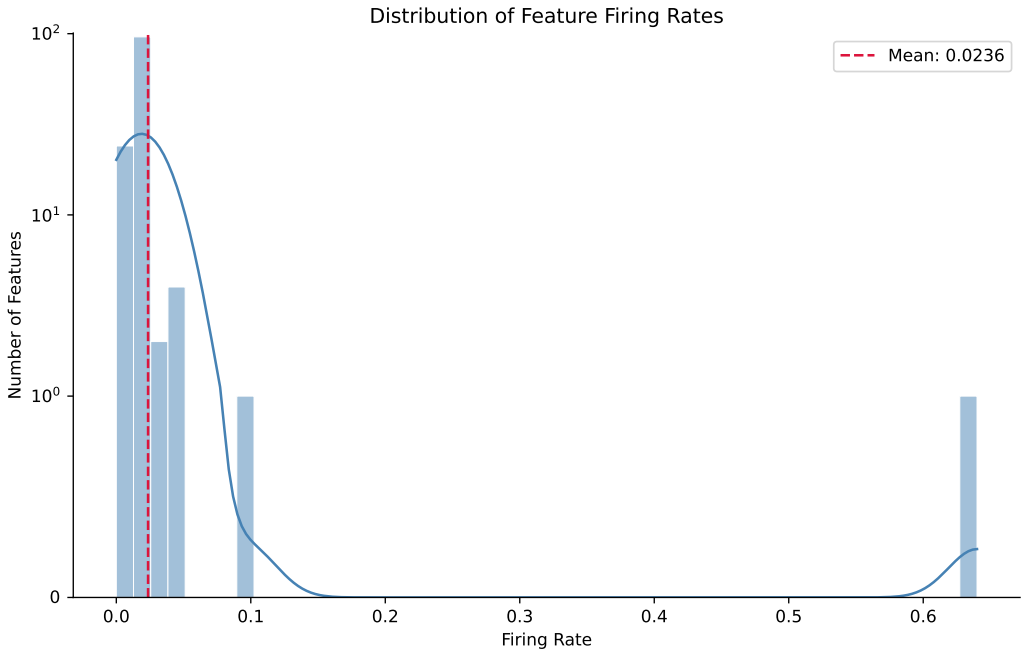

Figure 6. This histogram shows the distribution of firing rates for the SAE checkpoint's latents. The y-axis uses a symmetric logarithmic scale with a linear scale between $[-1, 1]$. A given latent's firing rate is the proportion of inputs which cause that latent to activate. The special latents are the only ones with a firing rate greater than one standard deviation from the mean (mean: 0.0236, standard deviation: 0.0559): latent 11 has rate 0.6400, latent 56 has rate 0.0991.

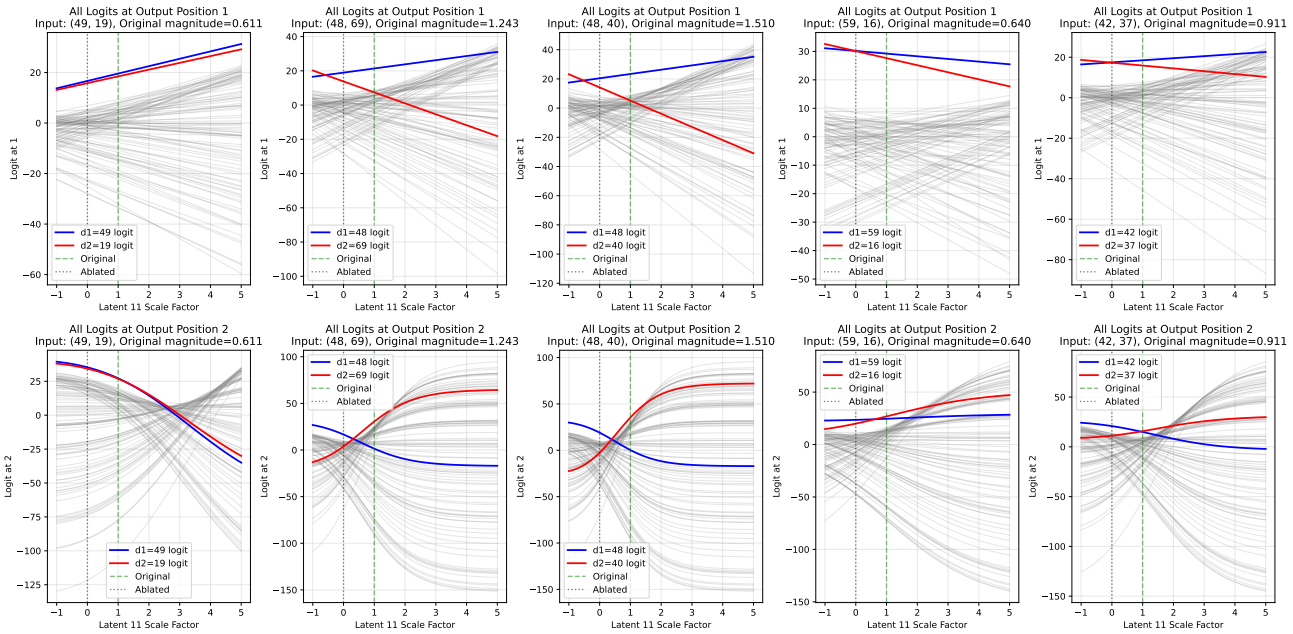

Figure 7. For 9.2% of inputs, latent 11 (in the BatchTopK SAE from Section **??**) must be negatively scaled to swap the predicted symbol at one of the output positions.

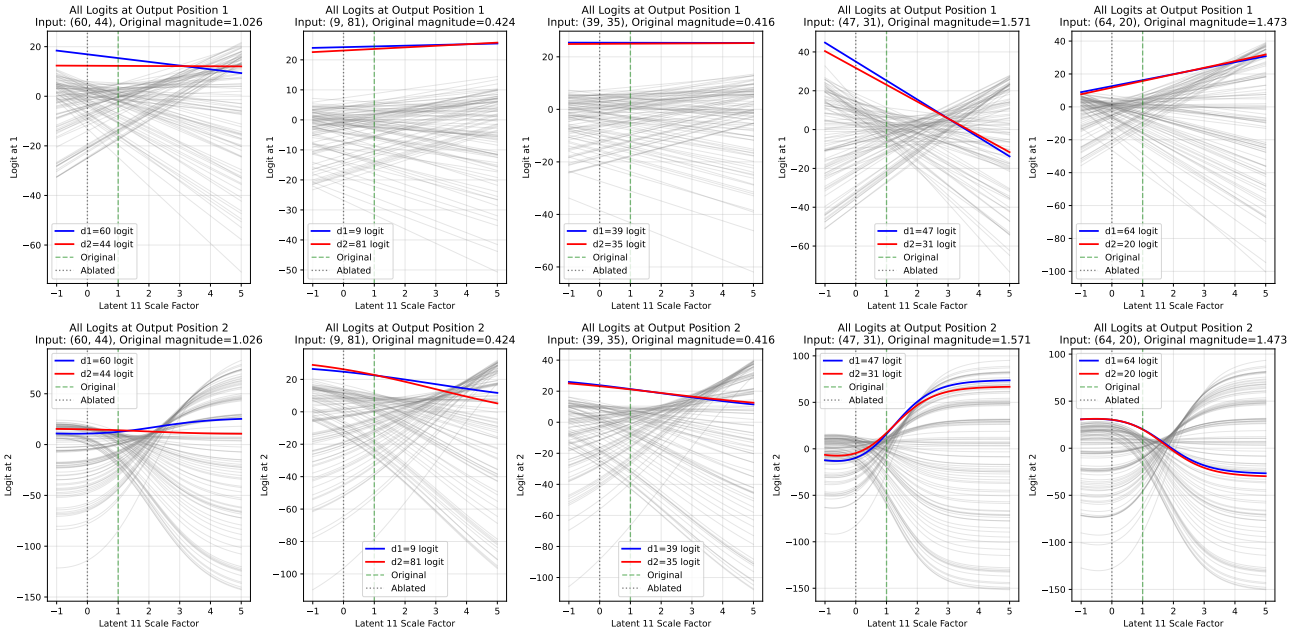

Figure 8. For 2.3% of inputs, it is possible to scale latent 11 (in the BatchTopK SAE from Section **??**) to swap the symbol at each output position individually but not simultaneously.

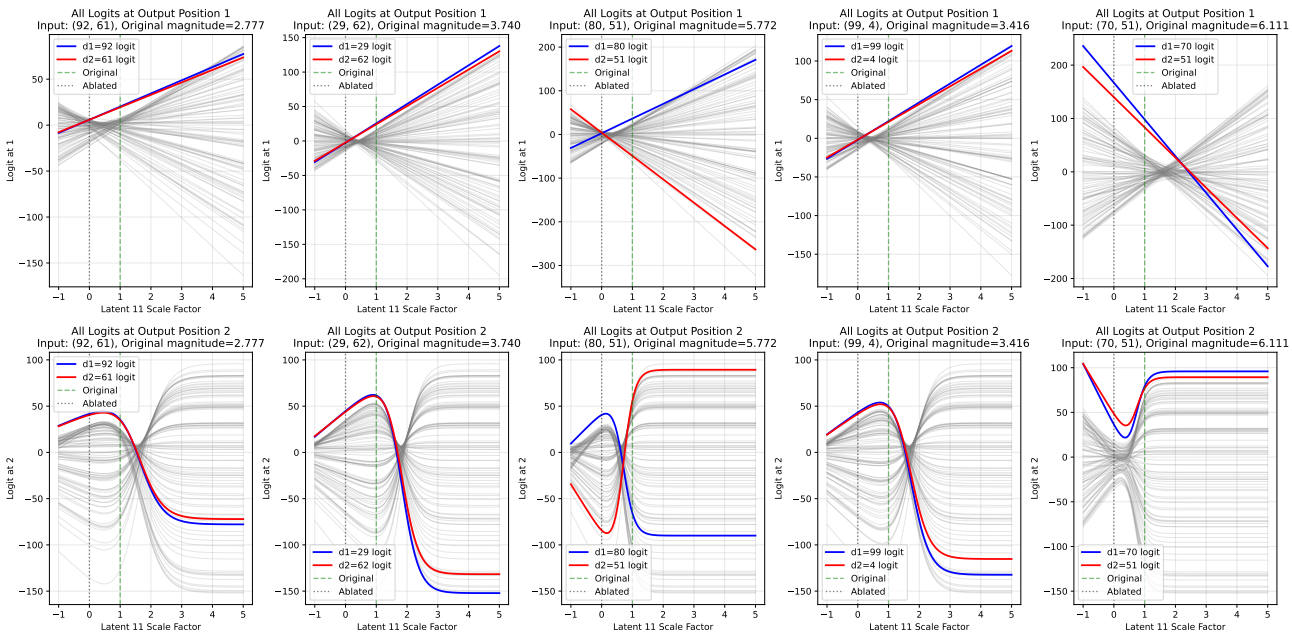

Figure 9. For 1.6% of inputs, the logit for the symbol at d_2 is always dominant at o_2 in the observed latent scale range (in the BatchTopK SAE from Section **??**), so there is no scale where the the symbol at d_1 is predicted instead.

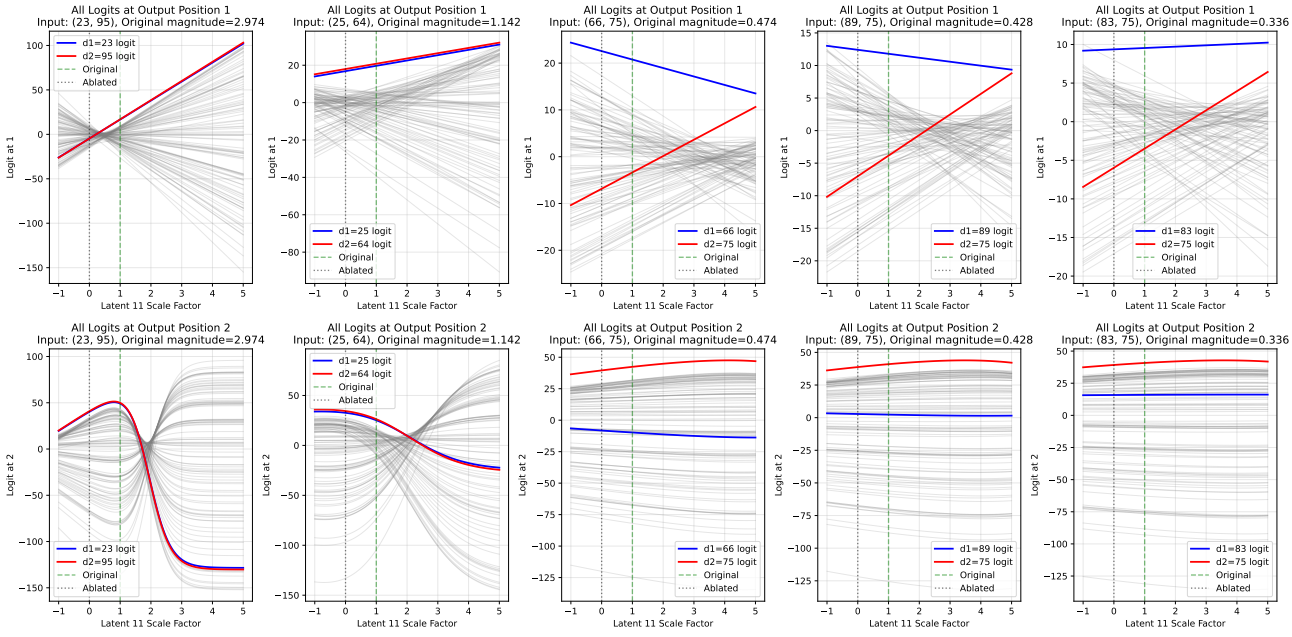

Figure 10. For 0.2% of inputs, the logit for the symbol at d_1 is always dominant at o_1 in the observed latent scale range (in the BatchTopK SAE from Section **??**), so there is no scale where the the symbol at d_2 is predicted instead.

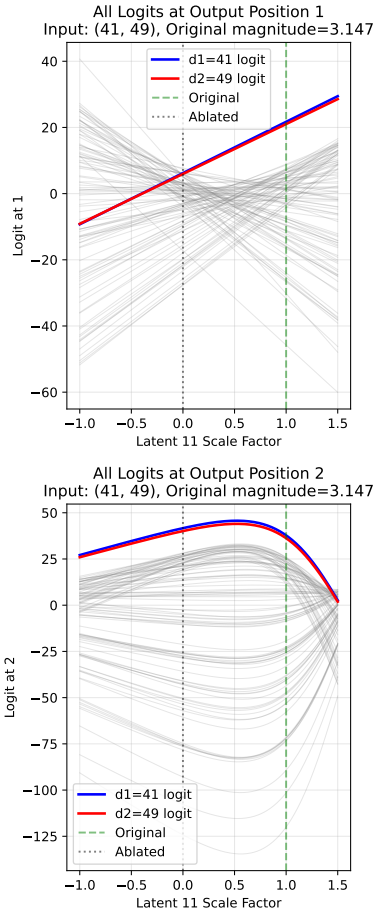

Figure 11. In the BatchTopK SAE from Section **??**, latent 11 cannot be scaled to swap input bigram $(41, 49)$ because the logit for 49 is never argmax in position o_1, as shown by the top plot.

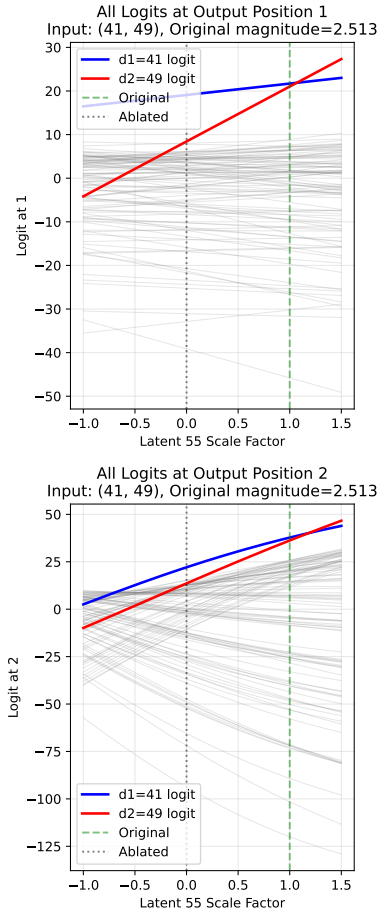

Figure 12. However, in the same SAE, latent 55 co-activates with latent 11 for input $(41, 49)$ and can be scaled by factor $p \in [1.07, 1.18]$ to cause the model to output $(49, 41)$ instead.

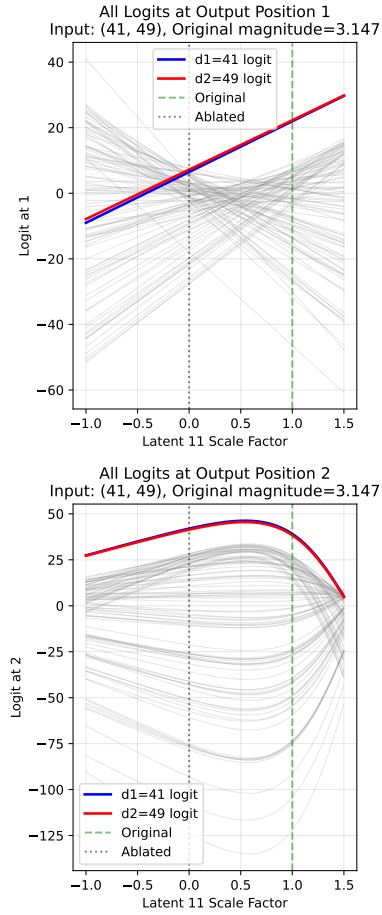

Figure 13. Moreover, in the same SAE, if latent 55's activation magnitude is scaled by a factor of 1.1, then we can now scale latent 11 by factor $p \in [0.03, 1.45]$ to swap the outputs, which was not possible without scaling latent 55. This suggests that some graded activation demonstrations are only possible when multiple active latents are scaled simultaneously.